数理分析方法

实数理论的基础定理

课件编号：000-ZSM-PPT　原始 PPT：62 页

逐页保留原始 PPT 画面 + 中文零基础自学讲解 + 证明专题

阅读说明：本讲义每一页原课件均以原始 PDF 页面完整嵌入；其后的中文解读旨在帮助初学者识别定义、假设、论证步骤。章末为从零展开的核心证明与复习提示。讲义属于学习辅助材料，不代替原教师课件。参考：Gérard Debreu (1959), *Theory of Value*。

目录

1 学前导读

这一讲学什么

本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。
实数完备性是多条定理背后的共同原因。做证明题时，尤其注意“非空、有界、闭、收敛”的区别。

证明题通用方法（先读本页）

严格证明的四个必要动作

第一：写出假设。比如“K 非空、闭、有界”，不能只写“K 很好”。
第二：逐字展开定义。写出 $\forall$（任意）、$\exists$（存在）、$\Rightarrow$（推出）的顺序。
第三：解释每一次定理调用。从有界得到收敛子列还缺 Bolzano–Weierstrass；得到极限在原集合还需要闭性。
第四：检查结论是否恰好是题目所求。“有极限”不等于“极限属于集合”，“函数有界”不等于“达到最大值”。

最常用的三种模板

集合相等：任取 x，证明 $x \in A \Rightarrow x \in B$，再证明 $x \in B \Rightarrow x \in A$。
极限：给定任意 $\varepsilon > 0$，找出一个 N（或 δ）并验证最终不等式。
存在性：常用反证、上确界构造、嵌套区间、连续函数极值、固定点定理；对每个方法要明确其全部前提。

第一张主线地图

有序域 + 上确界公理 $\Longrightarrow$ 阿基米德性、稠密性、闭区间套；闭区间套/完备性 $\Longrightarrow$ Bolzano–Weierstrass、Cauchy 准则；紧致性 + 连续性 $\Longrightarrow$ 最大值定理与介值定理等。证明中箭头两侧的假设一定写清。

2 逐页课件与中文讲解

2.1 第 1 页｜本页课件

Mathematical Analysis
Foundamental Theorems on Real Number Theory

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School of Finance
Renmin University of China

February 17, 2025

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

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2.2 第 2 页｜Real Number System

Real Number System

There are at least two approaches to defining the real number system.

Approach 1. Start with the set of natural numbers $\mathbb{N}$, construct the set $\mathbb{Z}$ of all integers $(\mathbb{N} \cup \{0\} \cup (-\mathbb{N}))$, and next, the set $\mathbb{Q}$ of rationals and then the set of $\mathbb{R}$ of real numbers either as the set of all Cauchy sequences of rationals or as Dedekind cuts. The step going from $\mathbb{Q}$ to $\mathbb{R}$ via Cauchy sequences is also available for completing any incomplete metric space.

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本页重点：有理数缺少某些极限点，例如逼近 $\sqrt{2}$ 的有理数数列。构造实数就是要填补这样的缺口。

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对应的完整证明：参见本讲义最后的「complete」证明专题。

2.3 第 3 页｜Real Number System

Real Number System

There are at least two approaches to defining the real number system.

Approach 2. Define the set of real numbers, $\mathbb{R}$, as a set that satisfies three sets of axioms. The first set is algebraic involving addition and multiplication. The second set is on ordering that, with the first, makes $\mathbb{R}$ an ordered field (Royden 1988). The third set is a single axiom known as the completeness axiom. Thus $\mathbb{R}$ is defined as complete ordered field.

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2.4 第 4 页｜Additive Group

Additive Group

A group is a set, G, together with an operation $+$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 + a_2$. To qualify as a group, the set and operation, $(G, +)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 + a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(3) Identity element: There exists an element 0 in G such that, for every element a in G, the equation $0 + a = a + 0 = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Additive Inverse: For each a in G, there exists an inverse element $-a$ in G such that $a + (-a) = (-a) + a = 0$.

A group $(G, +)$ with the property that $a_1 + a_2 = a_2 + a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

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2.5 第 5 页｜Multiplicative Group

Multiplicative Group

A group is a set, G, together with an operation $\times$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 \times a_2$. To qualify as a group, the set and operation, $(G, \times)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 \times a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(3) Identity element: There exists an element 1 in G such that, for every element a in G, the equation $1 \times a = a \times 1 = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Multiplicative Inverse: For each a in G, there exists an inverse element a^{-1} in G such that $a \times a^{-1} = a^{-1} \times a = 1$.

A group $(G, \times)$ with the property that $a_1 \times a_2 = a_2 \times a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

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2.6 第 6 页｜Group

Group

A group is a set, G, together with an operation $\cdot$ (called the group law of G) that combines any two elements a_1 and a_2 to form another element, denoted $a_1 \cdot a_2$. To qualify as a group, the set and operation, $(G, \cdot)$, must satisfy four requirements known as the group axioms:

(1) Closure: For all a_1 and a_2 in G, the result of the operation, $a_1 \cdot a_2$, is also in G.

(2) Associativity: For all a_1, a_2 and a_3 in G, $(a_1 \cdot a_2) \cdot a_3 = a_1 \cdot (a_2 \cdot a_3)$.

(3) Identity element: There exists an element e in G such that, for every element a in G, the equation $e \cdot a = a \cdot e = a$ holds. Such an element is unique, and thus one speaks of the identity element.

(4) Inverse element: For each a in G, there exists an inverse element a^{-1} in G such that $a \cdot a^{-1} = a^{-1} \cdot a = e$.

A group $(G, \cdot)$ with the property that $a_1 \cdot a_2 = a_2 \cdot a_1$ for all a_1 and a_2 in G is called Abelian (in honor of Niels Henrik Abel) or commutative. Groups not satisfying this property are said to be non-Abelian or non-commutative.

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2.7 第 7 页 | Ring

Ring

A ring is a set R equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) R is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all a_1 and a_2 in R, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in R.

(1.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in R such that, for every element a in R, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each a in R, there exists an inverse element $-a$ in R such that $a + (-a) = (-a) + a = 0$.

(2) R is a monoid under multiplication $\times$, meaning that:

(2.1) Closure: For all a_1 and a_2 in R, the result of the operation, $a_1 \times a_2$, is also in R.

(2.2) Associativity: For all a_1, a_2 and a_3 in R, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in R such that, for every element a in R, the equation $1 \times a = a \times 1 = a$ holds.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in R, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

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2.8 第 8 页｜Abelian Ring = Commutative Ring

Abelian Ring = Commutative Ring

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(1) R is an abelian group under addition $+$, meaning that:

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(1.3) Additive identity: There exists an element 0 in R such that, for every element a in R, the equation $0 + a = a + 0 = a$ holds.

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(2) R is a monoid under multiplication $\times$, meaning that:

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(2.3) Multiplicative Identity: There exists an element 1 in R such that, for every element a in R, the equation $1 \times a = a \times 1 = a$ holds.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in R, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

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2.9 第 9 页｜本页课件

Field

A field is a set F equipped with two binary operations $+$ and $\cdot$ satisfying the following three sets of axioms, called the ring axioms.

(1) F is an abelian group under addition $+$, meaning that:

(1.1) Commutativity: For all a_1 and a_2 in F, the result of the operation, $a_1 + a_2 = a_2 + a_1$, is also in F.

(1.2) Associativity: For all a_1, a_2 and a_3 in F, $(a_1 + a_2) + a_3 = a_1 + (a_2 + a_3)$.

(1.3) Additive identity: There exists an element 0 in F such that, for every element a in F, the equation $0 + a = a + 0 = a$ holds.

(1.4) Additive Inverse: For each a in F, there exists an inverse element $-a$ in F such that $a + (-a) = (-a) + a = 0$.

(2) F is a monoid under multiplication $\times$, meaning that:

(2.1) Commutativity: For all a_1 and a_2 in F, the result of the operation, $a_1 \times a_2 = a_2 \times a_1$, is also in F.

(2.2) Associativity: For all a_1, a_2 and a_3 in F, $(a_1 \times a_2) \times a_3 = a_1 \times (a_2 \times a_3)$.

(2.3) Multiplicative Identity: There exists an element 1 in F such that, for every element a in F, the equation $1 \times a = a \times 1 = a$ holds.

(2.4) Multiplicative Inverse: For each $a \neq 0$ in F, there exists an inverse element a^{-1} in F such that $a \times a^{-1} = a^{-1} \times a = 1$.

(3) Multiplication is distributive with respect to addition:

(3.1) Left Distributivity: For all a_1, a_2 and a_3 in F, $a_1 \times (a_2 + a_3) = a_1 \times a_2 + a_1 \times a_3$.

(3.2) Right Distributivity: For all a_1, a_2 and a_3 in R, $(a_2 + a_3) \times a_1 = a_2 \times a_1 + a_3 \times a_1$.

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2.10 第 10 页｜ Field

Field

If there is an element 1 in R such that $1 \neq 0$ and $1 \times a = a \times 1 = a$ for each element a in R, we say that R is a ring with unity or identity. A commutative ring R with identity is called an integral domain if, for every a_1 and a_2 in R such that $a_1 \times a_2 = 0$, either $a_1 = 0$ or $a_2 = 0$. A division ring is a ring R, with an identity, in which every non-zero element in R is a unit; that is, for each a in R with $a \neq 0$, there exists a unique element a^{-1} such that $a^{-1} \times a = a \times a^{-1} = 1$. A commutative division ring is called a field.

Rings: $\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

Fields: $\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$

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2.11 第 11 页｜Linear Space = Vector Space

Linear Space = Vector Space

A vector space over a field F is a set V together with two operations that satisfy the eight axioms listed below.

(1) The first operation, called vector addition or simply addition $+ : V \times V \to V$, takes any two vectors v_1 and v_2 and assigns to them a third vector which is commonly written as $v_1 + v_2$, and called the sum of these two vectors. (Note that the resultant vector is also an element of the set V).

(2) The second operation, called scalar multiplication $\cdot : F \times V \to V$, takes any scalar a and any vector v and gives another vector $a \cdot v$. (Similarly, the vector $a \cdot v$ is an element of the set V).

Elements of V are commonly called vectors. Elements of F are commonly called scalars.

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2.12 第 12 页｜Linear Space = Vector Space

Linear Space = Vector Space

(1) Commutativity: For all v_1 and v_2 in F, the result of the operation, $v_1 + v_2 = v_2 + v_1$, is also in V.

(2) Associativity: For all v_1, v_2 and v_3 in V, $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$.

(3) Additive identity: There exists an element 0 in V such that, for every element v in V, the equation $0 + v = v + 0 = v$ holds.

(4) Additive Inverse: For each v in V, there exists an inverse element $-v$ in V such that $v + (-v) = (-v) + v = 0$.

(5) Multiplicative Identity in Field F for Scalar Multiplication: For each v in V, $1 \cdot v = v$.

(6) Compatibility of Scalar Multiplication with Field Multiplication: For any a_1 and a_2 in F and each v in V, $a_1 \cdot (a_2 \cdot v) = (a_1 \times a_2) \cdot v$.

(7) Distributivity of Scalar Multiplication with respect to Vector Addition: For any a in F and each v_1 and v_2 in V, $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$.

(8) Distributivity of Scalar Multiplication with respect to Field Addition: For any a_1 and a_2 in F and each v in V, $(a_1 + a_2) \cdot v = a_1 \cdot v + a_2 \cdot v$.

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2.13 第 13 页｜Order Axioms

Order Axioms

Order Axioms There is a subset $\mathbb{P}$ of the set $\mathbb{R}$ of real numbers, called the set of strictly positive numbers such that:

(1) For any real number a in $\mathbb{R}$, exactly one of the following holds: $a = 0$ or $a \in \mathbb{P}$ or $-a \in \mathbb{P}$.

(2) If a_1 and a_2 in $\mathbb{P}$, $a_1 + a_2 \in \mathbb{P}$.

(3) If a_1 and a_2 in $\mathbb{P}$, $a_1 \cdot a_2 \in \mathbb{P}$.

The "Less Than" Relation We now define

- $a_1 < a_2$ to mean $a_2 - a_1 \in \mathbb{P}$ $(a_2 - a_1 \in \mathbb{R}_{++})$
- $a_1 \leqslant a_2$ to mean $a_2 - a_1 \in \mathbb{P}$ or $a_1 = a_2$ $(a_2 - a_1 \in \mathbb{R}_{+})$
- $a_1 > a_2$ to mean $a_1 - a_2 \in \mathbb{P}$ $(a_1 - a_2 \in \mathbb{R}_{++})$
- $a_1 \geqslant a_2$ to mean $a_1 - a_2 \in \mathbb{P}$ or $a_1 = a_2$ $(a_1 - a_2 \in \mathbb{R}_{+})$

$a_1 < a_2$ if and only if $a_2 > a_1$ and $a_1 \leqslant a_2$ if and only if $a_2 \geqslant a_1$.

$$\mathbb{P} = \mathbb{R}_{+} \setminus \{0\} = \mathbb{R}_{++}$$

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

结合原文理解：实数完备性是多条定理背后的共同原因。做证明题时，尤其注意“非空、有界、闭、收敛”的区别。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.14 第 14 页｜Ordered Field

Ordered Field

Any field F, together with two operations $\oplus$ and $\otimes$ and two members $0_\oplus$ and $0_\otimes$ of F, and a subset P of F, which satisfies the corresponding versions of (1) - (2) - (3), is called an ordered field.

Example. Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields.

这一页怎么读

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结合原文理解：实数完备性是多条定理背后的共同原因。做证明题时，尤其注意“非空、有界、闭、收敛”的区别。

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2.15 第 15 页｜本页课件

Completeness Axiom

Dedekind Completeness Axiom Suppose A and B are two (non-empty) sets of real numbers with the properties:

(1) if $a \in A$ and $b \in B$, then $a < b$.

(2) every real number is in either A or B (in symbols; $A \cup B = \mathbb{R}$).

Then there is a unique real number c such that:

(1) if $a < c$ then $a \in A$, and

(2) if $b > c$ then $b \in B$.

Note that every number strictly less than c belongs to A and every number strictly more than c belongs to B. Moreover, either $c \in A$ or $c \in B$ by (2). Hence if $c \in A$ then $A = (-\infty, c]$ and $B = (c, \infty)$; while if $c \in B$ then $A = (-\infty, c)$ and $B = [c, \infty)$.

The pair of sets $\{A, B\}$ is called a Dedekind Cut.

The intuitive idea of Dedekind Completeness Axiom is that there are no "holes" in the real numbers.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：有理数缺少某些极限点，例如逼近 $\sqrt{2}$ 的有理数数列。构造实数就是要填补这样的缺口。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成"任意…存在…"的中文句子，再试着写一个满足定义和一个不满足定义的例子。

对应的完整证明：参见本讲义最后的「complete」证明专题。

2.16 第 16 页｜Dedekind - Cantor Axiom of Continuity of a Real Line

Dedekind - Cantor Axiom of Continuity of a Real Line

Dedekind - Cantor Axiom of Continuity of a Real Line Suppose the pair of sets (A, A^c) is a Dedekind Cut of $\mathbb{R}$ with the property: if $a \in A$ and $b \in A^c$, then $a < b$. Then either there exists a maximum in A or there exists a minimum in A^c:

(1) If there exists a maximum c in A, then $A = (-\infty, c]$ and $A^c = (c, \infty)$.

(2) If there exists a minimum c in A^c, then $A = (-\infty, c)$ and $A^c = [c, \infty)$.

这一页怎么读

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读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「complete」证明专题。

2.17 第 17 页｜Supremum and Infimum

Supremum and Infimum

(1) Let S be a set of real numbers. u is an upper bound for S if $x \leqslant u$ for all $x \in S$. A set S is bounded above if it has an upper bound.

(2) b is the least upper bound (or l.u.b. or supremum or sup) for S if b is an upper bound, and moreover $b \leqslant u$ whenever u is any upper bound for S.

(1) Let S be a set of real numbers. d is an lower bound for S if $x \geqslant d$ for all $x \in S$. A set S is bounded below if it has a lower bound.

(2) l is the greatest lower bound (or g.l.b. or infimum or inf) for S if l is an lower bound, and moreover $l \geqslant d$ whenever d is any lower bound for S.

这一页怎么读

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读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.18 第 18 页｜Supremum and Infimum Completeness Axiom

Supremum and Infimum Completeness Axiom

Theorem of Supremum (Least Upper Bound Completeness Axiom)
Suppose S is a non-empty set of real numbers which is bounded above. Then S has a supremum in $\mathbb{R}$.

Theorem of Infimum (Greatest Lower Bound Completeness Axiom)
Suppose S is a non-empty set of real numbers which is bounded below. Then S has a infimum in $\mathbb{R}$.

这一页怎么读

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读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.19 第 19 页｜本页课件

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded above. Let

$$B = \{x \in \mathbb{R} : x \text{ is an upper bound for } S\} \quad \text{and} \quad A = \mathbb{R} \setminus B.$$

Note that $B \neq \emptyset$; and if $x \in S$ then $x - 1$ is not an upper bound for S so $A \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then a would also be an upper bound for S, which contradicts the definition of A.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \sup S$.

If $c \in A$ then c is not an upper bound for S and so there exists $x \in S$ with $c < x$. But then $a = \dfrac{c + x}{2}$ is not an upper bound for S, i.e. $a \in A$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $a \leqslant c$ for all $a \in A$. Hence $c \in B$. But if $c \in B$ then $c \leqslant b$ for all $b \in B$; i.e. c is less than any upper bound for S.

这一页怎么读

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对应的完整证明：参见本讲义最后的「complete」证明专题。

2.20 第 20 页｜ Dedekind - Cantor Axiom vs Supremum and Infimum Complet

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then A is bounded above (by any element of B). Let $c = \sup A$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Now every member of B is an upper bound for A, from the first property of a Dedekind cut; hence $a \notin B$, as otherwise a would be an upper bound for A which is less than the least upper bound c. Hence $a \in A$.
- Next suppose $b > c$. Since c is an upper bound for A (in fact the least upper bound), it follows we cannot have $b \in A$, and thus $b \in B$.

这一页怎么读

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读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.21 第 21 页｜本页课件

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Necassity] Suppose that S is a non-empty set of real numbers which is bounded below. Let

$$A = \{x \in \mathbb{R} : x \text{ is a lower bound for } S\} \quad \text{and} \quad B = \mathbb{R} \setminus A.$$

Note that $A \neq \emptyset$; and if $x \in S$ then $x + 1$ is not a lower bound for S so $B \neq \emptyset$.

Suppose $a \in A$ and $b \in B$, then $a < b$. If $a \geqslant b$, then b would also be a lower bound for S, which contradicts the definition of B.

Let c be the real number given by Dedekind - Cantor Axiom. We claim that $c = \inf S$.

If $c \in B$ then c is not a lower bound for S and so there exists $x \in S$ with $x < c$. But then $b = \dfrac{x + c}{2}$ is not a lower bound for S, i.e. $b \in B$, contradicting the fact from the conclusion of Dedekind - Cantor Axiom that $c \leqslant b$ for all $b \in B$. Hence $c \in A$. But if $c \in A$ then $a \leqslant c$ for all $a \in A$; i.e. c is more than any lower bound for S.

这一页怎么读

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对应的完整证明：参见本讲义最后的「complete」证明专题。

2.22 第 22 页｜ Dedekind - Cantor Axiom vs Supremum and Infimum Complet

Dedekind - Cantor Axiom vs Supremum and Infimum Completeness Axiom

[Sufficiency] For this, suppose $\{A, B\}$ is a Dedekind cut. Then B is bounded below (by any element of A). Let $c = \inf B$. We claim that

$$a < c \Rightarrow a \in A, \qquad \text{and} \qquad b > c \Rightarrow b \in B.$$

- Suppose $a < c$. Since c is a lower bound for B (in fact the greatest lower bound), it follows we cannot have $a \in B$, and thus $a \in A$.
- Next suppose $c < b$. Now every member of A is a lower bound for B, from the first property of a Dedekind cut; hence $b \notin A$, as otherwise b would be a lower bound for B which is less than the greatest lower bound c. Hence $b \in B$.

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.23 第 23 页｜Proposition on Supremum and Infimum

Proposition on Supremum and Infimum

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.24 第 24 页｜Definition of Supremum

Definition of Supremum

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2) $b \leqslant u$ whenever u is any upper bound for S.

Suppose S is a non-empty set of real numbers. Then $b = \sup S$ iff

(1) $x \leqslant b$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x > b - \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \leqslant b - \varepsilon$ for every $x \in S$, i.e. $b - \varepsilon$ is an upper bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $u < b$ then from $(2')$ it follows that u is not an upper bound of S. That is (2): $b \leqslant u$ whenever u is any upper bound for S.

这一页怎么读

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.25 第 25 页｜Definition of Infimum

Definition of Infimum

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2) $l \geqslant d$ whenever d is any lower bound for S.

Suppose S is a non-empty set of real numbers. Then $l = \inf S$ iff

(1) $x \geqslant l$ for all $x \in S$;

(2′) for each $\varepsilon > 0$ there exists $x \in S$ such that $x < l + \varepsilon$.

$[(1) + (2) \Rightarrow (2')]$ Suppose $(2')$ is not true. Then for some $\varepsilon > 0$ it follows that $x \geqslant l + \varepsilon$ for every $x \in S$, i.e. $l + \varepsilon$ is a lower bound for S. This is a contradiction.

$[(1) + (2') \Rightarrow (2)]$ If $d > l$ then from $(2')$ it follows that d is not a lower bound of S. That is (2): $l \geqslant d$ whenever d is any lower bound for S.

这一页怎么读

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本页重点：下确界是所有下界中最大的一个。最好先画数轴，再将量词写成 a A 与 ε>0。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

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2.26 第 26 页｜Bounded Monotone Sequence

Bounded Monotone Sequence

Definition. A sequence is bounded if the set of terms from the sequence is bounded.

Definition. A sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is

(1) increasing if $x_n \leqslant x_{n+1}$ for all $n \in \mathbb{N}$;

(2) decreasing if $x_n \geqslant x_{n+1}$ for all $n \in \mathbb{N}$;

(3) strictly increasing if $x_n < x_{n+1}$ for all $n \in \mathbb{N}$;

(4) strictly decreasing if $x_n > x_{n+1}$ for all $n \in \mathbb{N}$.

A sequence is monotone if it is either increasing or decreasing.

这一页怎么读

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2.27 第 27 页｜Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence Every bounded monotone sequence in $\mathbb{R}$ has a limit in $\mathbb{R}$.

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

这一页怎么读

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2.28 第 28 页｜Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is increasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded above, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \leqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N > x - \varepsilon$, as otherwise $x - \varepsilon$ would be an upper bound. Choose such $N = N(\varepsilon)$. Since $x_N > x - \varepsilon$, then $x_n > x - \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x - \varepsilon < x_n \leqslant x$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

这一页怎么读

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读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.29 第 29 页｜Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Proof Suppose $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ and $\{x_n : n \in \mathbb{N}\}$ is decreasing. Since the set of terms $\{x_n : n \in \mathbb{N}\}$ is bounded below, it has a supremum x. We claim that $\lim_{n\to\infty} x_n = x$.

To see this, note that $x_n \geqslant x$ for all $n \in \mathbb{N}$; but if $\varepsilon > 0$ then there exists $N \in \mathbb{N}$ such that $x_N < x + \varepsilon$, as otherwise $x + \varepsilon$ would be a lower bound. Choose such $N = N(\varepsilon)$. Since $x_N < x + \varepsilon$, then $x_n < x + \varepsilon$ for all $n \geqslant N$ as the sequence is increasing. Hence

$$x \leqslant x_n < x + \varepsilon$$

for all $n \geqslant N$. Thus $|x - x_n| < \varepsilon$ for $n \geqslant N$, and so $\lim_{n\to\infty} x_n = x$ (since $\varepsilon > 0$ is arbitrary).

这一页怎么读

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本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.30 第 30 页｜Cantor Nested Intervals Theorem

Cantor Nested Intervals Theorem

Theorem 1 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = \sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = \sup\{d_n : n \in \mathbb{N}\}$ and $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

$\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals if $I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq \cdots$.

这一页怎么读

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对应的完整证明：参见本讲义最后的「nested」证明专题。

2.31 第 31 页｜Cantor Nested Intervals Theorem

Cantor Nested Intervals Theorem

Theorem 1 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $D = \sup\{d_n : n \in \mathbb{N}\}$ such that $D \in \bigcap_{n=1}^{\infty} I_n$.

Proof We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_n \leqslant u_1$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $A = \{d_n : n \in \mathbb{N}\}$ that is bounded above by u_1. Thus this set has a supremum in the real numbers and denote it $D = \sup A = \lim_{n\to\infty} d_n$ so that $d_n \leqslant D$ for all $n \in \mathbb{N}$.
We now want to show that $D \leqslant u_n$ for all $n \in \mathbb{N}$. We will do this by showing that $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_n \leqslant u_{n'}$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_n \leqslant u_{n'}$.
So then $d_n \leqslant u_{n'}$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $u_{n'}$ is an upper bound to the set A and so $D = \sup A \leqslant u_{n'}$ for all $n' \in \mathbb{N}$. Furthermore we have that $d_{n'} \leqslant D$ for all $n' \in \mathbb{N}$, and so $D \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $D \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.32 第 32 页｜Cantor Nested Intervals Theorem

Cantor Nested Intervals Theorem

Theorem 2 If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exists a real number $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $U \in \bigcap_{n=1}^{\infty} I_n$.

Proof We note that by the definition of nested intervals that $I_n \subseteq I_1$ for all $n \in \mathbb{N}$ so then $d_1 \leqslant u_n$ for all $n \in \mathbb{N}$.
Now consider the non-empty set $B = \{u_n : n \in \mathbb{N}\}$ that is bounded above by d_1. Thus this set has an infumum in the real numbers and denote it $U = \inf B = \lim_{n\to\infty} u_n$ so that $U \leqslant u_n$ for all $n \in \mathbb{N}$.
We now want to show that $I_n \leqslant U$ for all $n \in \mathbb{N}$. We will do this by showing that $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$.
First consider the case where $n \leqslant n'$. We thus have that $I_n \supseteq I_{n'}$ by the definition of nested intervals and so we get that $d_n \leqslant d_{n'} \leqslant u_{n'} \leqslant u_n$. Here we see that $d_{n'} \leqslant u_n$.
Now consider the case where $n > n'$. We thus have that $I_{n'} \supseteq I_n$ by the definition of nested intervals and so we have that $d_{n'} \leqslant d_n \leqslant u_n \leqslant u_{n'}$. Once again we have that $d_{n'} \leqslant u_n$.
So then $d_{n'} \leqslant u_n$ for all $n \in \mathbb{N}$ and $n' \in \mathbb{N}$, and so $I_{n'}$ is a lower bound to the set B and so $u_{n'} \leqslant \sup B = U$ for all $n' \in \mathbb{N}$. Furthermore we have that $U \leqslant u_{n'}$ for all $n' \in \mathbb{N}$, and so $U \in I_{n'}$ for every $n' \in \mathbb{N}$ and thus $U \in \bigcap_{n=1}^{\infty} I_n$ and so the set theoretic union is non-empty.

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对应的完整证明：参见本讲义最后的「nested」证明专题。

2.33 第 33 页｜本页课件

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = \sup\{d_n : n \in \mathbb{N}\} = \lim_{n\to\infty} d_n$ and $U = \inf\{u_n : n \in \mathbb{N}\} = \lim_{n\to\infty} u_n$ such that $[D, U] = \bigcap_{n=1}^{\infty} I_n$.

Proof We first prove that $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$. Now let $x \in [D, U]$, then $D \leqslant x \leqslant U$. But we know that $d_n \leqslant D$ for all $n \in \mathbb{N}$ and $U \leqslant u_n$ for all $n \in \mathbb{N}$ and so $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Therefore $x \in [d_n, u_n]$ for all $n \in \mathbb{N}$ or in other words $x \in \bigcap_{n=1}^{\infty} I_n$. Therefore $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$.

We then prove that $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$. Let $x \in \bigcap_{n=1}^{\infty} I_n$. Then $d_n \leqslant x \leqslant u_n$ for all $n \in \mathbb{N}$. We also know that $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$. Suppose that x is such that $d_n \leqslant x < D$. Since D is the supremum of the set A then there exists an element $d_x \in \{d_n : n \in \mathbb{N}\}$ such that $x < d_x$ and so $x \notin [d_x, u_x]$ and therefore $x \notin \bigcap_{n=1}^{\infty} I_n$, a contradiction. Now suppose that x is such that $U < x \leqslant u_n$ for all $n \in \mathbb{N}$. Since U is the infimum of the set B then there exists an element $u_x \in \{u_n : n \in \mathbb{N}\}$ such that $u_x < x$ and so $x \notin [d_x, u_x]$ and so $x \notin \bigcap_{n=1}^{\infty} I_n$, once again, a contradiction. We must therefore have that $d_n \leqslant D \leqslant x \leqslant U \leqslant u_n$ and so $x \in [D, U]$ and so $\bigcap_{n=1}^{\infty} I_n \subseteq [D, U]$.

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.34 第 34 页｜Cantor Nested Intervals Theorem

Cantor Nested Intervals Theorem

Nested Intervals Theorem If $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals and $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $\lim_{n\to\infty} d_n = \lim_{n\to\infty} u_n \in \bigcap_{n=1}^{\infty} I_n$.

Proof The non-empty set $A = \{d_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above by u_1, then it has a limit $D = \lim_{n\to\infty} d_n$.

The non-empty set $B = \{u_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below by d_1, then it has a limit $U = \lim_{n\to\infty} u_n$.

Thus $d_n \leqslant D \leqslant U \leqslant u_n$ for all $n \in \mathbb{N}$, and

$$0 \leqslant U - D \leqslant u_n - d_n.$$

Note $\lim_{n\to\infty} |I_n| = \lim_{n\to\infty} (u_n - d_n) = 0$, then $D = U \in \bigcap_{n=1}^{\infty} I_n$.

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对应的完整证明：参见本讲义最后的「nested」证明专题。

2.35 第 35 页｜本页课件

Bolzano Theorem

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a limit point.

Proof Let $\{x_n : n \in \mathbb{N}\}$ be a bounded sequence, we assume $\{x_n : n \in \mathbb{N}\} \subseteq [D, U]$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_1, U_1]$, and we take $x_{n_1} \in [D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts $\left[D_1, \frac{D_1+U_1}{2}\right]$ and $\left[\frac{D_1+U_1}{2}, U_1\right]$. There exists an interval including infinite items of sequence $\{x_n : n \in \mathbb{N}\}$, this interval is written as $[D_2, U_2]$, and we take $x_{n_2} \in [D_2, U_2]$ with $n_2 > n_1$. We repeat this step and obtain a sequence $\{[D_k, U_k] : k \in \mathbb{N}\}$ of closed bounded nested intervals:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_k, U_k] \supseteq \cdots \quad \text{with } U_k - D_k = \frac{U - D}{2^k}$$

with a subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $x_{n_k} \in [D_k, U_k]$ for $k \in \mathbb{N}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{k=1}^{\infty} [D_k, U_k]$, and $|x_{n_k} - a| < U_k - D_k = \frac{U-D}{2^k}$, then $\lim_{k \to \infty} x_{n_k} = a$.

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对应的完整证明：参见本讲义最后的「nested」证明专题。

2.36 第 36 页｜本页课件

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Proof. Note that E is a bounded infinite set in $\mathbb{R}$, there exists a constant M such that, for any $x \in E$, $|x| \leqslant M$. Define $I_0 = [-M, M]$, we divide R_0 into two small closed intervals: $[-M, 0]$ and $[0, M]$, then at least one small interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_1 with length M. Generally, we assume that there exists a sequence of closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_N, \quad I_N \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots, N$. We divide I_N into two small closed intervals, then at least one small closed interval includes infinite points in E (otherwise E is a finite set!), we call such a small closed interval I_{N+1} with length $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed intervals

$$I_0 \supseteq I_1 \supseteq \cdots \supseteq I_n \supseteq \cdots, \quad I_n \cap E \text{ is an infinite set}$$

where the length of I_n is $\frac{M}{2^{n-1}}$ for $n = 1, 2, \cdots$.

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读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.37 第 37 页｜Weierstrass Theorem on Accumulation Point

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.

Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $x \in \bigcap_{n=1}^{\infty} I_n$. We now check that this x is one of accumulation points of E.
Note $\lim_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large integer n_0, $\frac{M}{2^{n_0}} < \delta$. The length of I_{n_0+1} is strictly less than δ, then $I_{n_0+1} \subseteq N(x,\delta)$, and hence $E \cap I_{n_0+1} \subseteq E \cap N(x,\delta)$. x is an accumulation point of E.

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对应的完整证明：参见本讲义最后的「nested」证明专题。

2.38 第 38 页｜Cauchy Convergence Criterion

Cauchy Convergence Criterion

Definition. $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is a Cauchy sequence if for any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$.

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

这一页怎么读

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本页重点：柯西条件只比较尾部的各项，而不预设极限；其充分性使用实数的完备性。

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对应的完整证明：参见本讲义最后的「cauchy」证明专题。

2.39 第 39 页｜ Cauchy Convergence Criterion

Cauchy Convergence Criterion

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ is convergent if and only if it is a Cauchy sequence.

Proof [Necessity] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a convergent sequence, and suppose $x = \lim_{n\to\infty} x_n$. For any $\varepsilon > 0$ there exists an integer $N \in \mathbb{N}$ such that, when $n > N$, $|x_n - x| < \varepsilon$. It follows that for any $n_1 > N$ and $n_2 > N$,

$$|x_{n_1} - x_{n_2}| \leqslant |x_{n_1} - x| + |x_{n_2} - x| < 2\varepsilon.$$

Thus $\{x_n : n \in \mathbb{N}\}$ is a Cauchy sequence.

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对应的完整证明：参见本讲义最后的「cauchy」证明专题。

2.40 第 40 页｜本页课件

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|x_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

From Bolzano-Weierstrass Theorem, the bounded sequence $\{x_n : n \in \mathbb{N}\}$ has a convergent subsequence $\{x_{n_k} : k \in \mathbb{N}\}$ with $\lim\limits_{k\to\infty} x_{n_k} = a$. For any $\varepsilon > 0$ there exists an integer K such that, when $k > K$, $|x_{n_k} - a| < \varepsilon$.
Take $k_0 = \max\{K+1, N+1\}$, then $n_{k_0} \geqslant n_{N+1} \geqslant N+1 > N$.
Then, when $n > N$, $|x_n - x_{n_{k_0}}| < \varepsilon$, and hence

$$|x_n - a| \leqslant |x_n - x_{n_{k_0}}| + |x_{n_{k_0}} - a| < 2\varepsilon$$

that is, $\lim\limits_{n\to\infty} x_n = a$.

这一页怎么读

概念主线： 本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点： 有界数列存在收敛子列。反复把区间一分为二，始终留下包含无穷多项的半区间。

读课件时的动作： 本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明： 参见本讲义最后的「bw」证明专题。

2.41 第 41 页｜本页课件

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \inf\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \geqslant y_n$ since y_{n+1} is the infimum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is an increasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：下确界是所有下界中最大的一个。最好先画数轴，再将量词写成 a A 与 ε>0。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.42 第 42 页｜本页课件

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \leqslant x_n$ for each $n \in \mathbb{N}$, then $y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon$, then
$x_{n_2} \leqslant \inf\{x_{n_1} + \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} = \inf\{x_{n_1} : n_1 \geqslant N_2\} + \frac{1}{2}\varepsilon = y_{n_1} + \frac{1}{2}\varepsilon$.
Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon \leqslant x_{n_2} < y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim\limits_{n\to\infty} x_n = a$.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

结合原文理解：实数完备性是多条定理背后的共同原因。做证明题时，尤其注意“非空、有界、闭、收敛”的区别。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.43 第 43 页｜本页课件

Cauchy Convergence Criterion

[Sufficiency] Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$. Then, for any $n > N$, $|x_n - x_{N+1}| < \varepsilon$, or

$$|n_n| \leqslant |x_n - x_{N+1}| + |x_{N+1}| < |x_{N+1}| + \varepsilon \leqslant M$$

where $M = \max\{|x_n| : n = 1, \cdots, N+1\} + \varepsilon$, thus $\{x_n : n \in \mathbb{N}\}$ is bounded.

Define $y_n = \sup\{x_n, x_{n+1}, \cdots\}$ for each $n \in \mathbb{N}$. It follows that $y_{n+1} \leqslant y_n$ since y_{n+1} is the supremum over a subset of the set corresponding to y_n, $\{y_n : n \in \mathbb{N}\}$ is a decreasing sequence. Moreover the sequence $\{y_n : n \in \mathbb{N}\}$ is bounded since the sequence $\{x_n : n \in \mathbb{N}\}$ is Cauchy and hence bounded. From Theorem on monotone sequences, there exists $a \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} y_n = a$. For any $\varepsilon > 0$ there exists an integer N_1 such that, when $n > N_1$, $|y_n - a| < \frac{1}{2}\varepsilon$, or $a - \frac{1}{2}\varepsilon < y_n < a + \frac{1}{2}\varepsilon$. We claim that $\lim\limits_{n\to\infty} x_n = a$.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.44 第 44 页｜本页课件

Cauchy Convergence Criterion

Let $\{x_n : n \in \mathbb{N}\} \subseteq \mathbb{R}$ be a Cauchy sequence. For any $\varepsilon > 0$ there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$, $|x_{n_1} - x_{n_2}| < \frac{1}{2}\varepsilon$, or

$$x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon.$$

From the definition of sequence $\{y_n : n \in \mathbb{N}\}$, $y_n \geqslant x_n$ for each $n \in \mathbb{N}$, then $x_{n_2} < x_{n_1} + \frac{1}{2}\varepsilon \leqslant y_{n_1} + \frac{1}{2}\varepsilon$ for $n_1 > N_1$ and $n_2 > N_2$. On the other hand, $x_{n_1} - \frac{1}{2}\varepsilon < x_{n_2}$, then $y_{n_1} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} : n_1 \geqslant N_2\} - \frac{1}{2}\varepsilon = \sup\{x_{n_1} - \frac{1}{2}\varepsilon : n_1 \geqslant N_2\} \leqslant x_{n_2}$. Thus there exists an integer N_2 such that, when $n_1 > N_2$ and $n_2 > N_2$,

$$y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon.$$

Take $N = \max\{N_1, N_2\}$, then, when $n_1 > N$ and $n_2 > N$,

$$a - \varepsilon < y_{n_1} - \frac{1}{2}\varepsilon < x_{n_2} \leqslant y_{n_1} + \frac{1}{2}\varepsilon < a + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, it follows that $\lim_{n\to\infty} x_n = a$.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

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2.45 第 45 页｜Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem on Finite Coverings

A collection $\mathscr{H}$ of open sets is an open covering of a set S if every point in S is contained in a set H belonging to $\mathscr{H}$; that is, $S \subseteq \{H \in \mathscr{H}\}$.

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：开覆盖具有有限子覆盖的结论必须依赖紧致性。证明时可用闭区间套反证或逐次细分。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「hb」证明专题。

2.46 第 46 页｜Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Proof (Reduction to Absurdity) Let $\mathscr{H}$ be an open covering of $[D, U]$. Assume that $[D, U]$ can not be covered by finite many open sets belonging to $\mathscr{H}$. We divide closed interval $[D, U]$ into two equal-length parts $\left[D, \frac{D+U}{2}\right]$ and $\left[\frac{D+U}{2}, U\right]$. There exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_1, U_1]$. Then we divide closed interval $[D_1, U_1]$ into two equal-length parts, then there exists at least one small interval that can not be covered by finite many open sets belonging to $\mathscr{H}$, this interval is written as $[D_2, U_2]$. We repeat this step and obtain a sequence $\{[D_n, U_n] : n \in \mathbb{N}\}$ of closed bounded nested intervals that can not be covered by finite many open sets belonging to $\mathscr{H}$:

$$[D_1, U_1] \supseteq [D_2, U_2] \supseteq \cdots \supseteq [D_n, U_n] \supseteq \cdots \quad \text{with } U_n - D_n = \frac{U - D}{2^n}.$$

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.47 第 47 页｜Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset S of the real line; then S has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

From Cantor Nested Intervals Theorem there exists a unique point $a \in \bigcap_{n=1}^{\infty} [D_n, U_n]$. Note that $\mathscr{H}$ is an open covering of $[D, U]$, then there exists $H_0 \in \mathscr{H}$ such that $a \in H_0$. Note that H_0 is an open set, then there exists $\delta > 0$ such that $(x-\delta, x+\delta) \subseteq H_0$.
Note $\lim_{n\to\infty} D_n = a = \lim_{n\to\infty} U_n$, then there exists an integer N such that, when $n > N$, $D_n \in (x-\delta, x+\delta)$ and $U_n \in (x-\delta, x+\delta)$.

$$[D_n, U_n] \subseteq (x-\delta, x+\delta) \subseteq H_0 \qquad \text{for } n > N.$$

which contradicts that $[D_n, U_n]$ can not be covered by finite many open sets belonging to $\mathscr{H}$.

这一页怎么读

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2.48 第 48 页｜Real Number Space R

Real Number Space $\mathbb{R}$

Dedekind - Cantor Axiom of Continuity of a Real Line Suppose the pair of sets (A, A^c) is a Dedekind Cut of $\mathbb{R}$ with the property: if $a \in A$ and $b \in A^c$, then $a < b$, then either there exists a maximum in A or there exists a minimum in A^c.

Theorem of Supremum and Infimum Suppose E is a non-empty set of real numbers which is bounded above, then E has a supremum in $\mathbb{R}$. Suppose E is a non-empty set of real numbers which is bounded below, then E has a infimum in $\mathbb{R}$.

Theorem of Limit of Monotone Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above, then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists. Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below, then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Nested Intervals Theorem If the interval $\{I_n = [d_n, u_n] : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals, then there exist two real numbers $D = \sup\{d_n : n \in \mathbb{N}\}$ and $U = \inf\{u_n : n \in \mathbb{N}\}$ such that $[D, U] \subseteq \bigcap_{n=1}^{\infty} I_n$.

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.49 第 49 页｜Real Number Space R

Real Number Space $\mathbb{R}$

Bolzano Theorem Each bounded sequence in $\mathbb{R}$ has a convergent subsequence.
Weierstrass Theorem Every bounded infinite set in $\mathbb{R}$ has an accumulation point.
Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\}$ in $\mathbb{R}$ is convergent if and only if it is a Cauchy sequence: for any $\varepsilon > 0$, there exist a large natural number N such that, when $n_1 > N$ and $n_2 > N$, $|x_{n_1} - x_{n_2}| < \varepsilon$.
Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset E of the real line; then E has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

这一页怎么读

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对应的完整证明：参见本讲义最后的「bw」证明专题。

2.50 第 50 页｜Propositions of Continuous Function on Closed Interval

Propositions of Continuous Function on Closed Interval

Theorem on Boundedness If function f is continuous on a bounded closed interval, then it is bounded.

Hint: Bolzano-Weierstrass Theorem

Hint: Heine - Borel Theorem on Finite Coverings

Theorem on Maximum and Minimum Values If function f is continuous on a bounded closed interval, then it has (reaches) maximum and minimum values.

Hint: Theorem of Supremum and Infimum + Bolzano-Weierstrass Theorem

Theorem on Solution If function f is continuous on a bounded closed interval $[D, U]$, and $f(D)f(U) < 0$, then there exists a solution on (D, U) to equation $f(x) = 0$.

Hint: Cantor Nested Intervals Theorem

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对应的完整证明：参见本讲义最后的「supremum」证明专题。

2.51 第 51 页｜Propositions of Continuous Function on Closed Interval

Propositions of Continuous Function on Closed Interval

Theorem on Inverse Function If function $y = f(x)$ is continuous and strictly monotonic increasing (or decreasing) on a bounded closed interval $[D, U]$, and $f(D) = L$ and $f(U) = B$, then there exists a continuous and strictly monotonic increasing (or decreasing) inverse function $x = g(y)$ on bounded closed interval $[L, B]$ (or $[B, L]$).

Cantor Theorem on Uniform Continuity If function f is continuous on a bounded closed interval, then it is uniformly continuous.

Hint: Bolzano-Weierstrass Theorem

Hint: Heine - Borel Theorem on Finite Coverings

这一页怎么读

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对应的完整证明：参见本讲义最后的「nested」证明专题。

2.52 第 52 页｜Linear Space = Vector Space

Linear Space = Vector Space

A vector space over a field F is a set V together with two operations that satisfy the eight axioms listed below.

(1) The first operation, called vector addition or simply addition $+ : V \times V \to V$, takes any two vectors v_1 and v_2 and assigns to them a third vector which is commonly written as $v_1 + v_2$, and called the sum of these two vectors. (Note that the resultant vector is also an element of the set V).

(2) The second operation, called scalar multiplication $\cdot : F \times V \to V$, takes any scalar a and any vector v and gives another vector $a \cdot v$. (Similarly, the vector $a \cdot v$ is an element of the set V).

Elements of V are commonly called vectors. Elements of F are commonly called scalars.

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概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

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2.53 第 53 页｜Linear Space = Vector Space

Linear Space = Vector Space

(1) Commutativity: For all v_1 and v_2 in V, the result of the operation, $v_1 + v_2 = v_2 + v_1$, is also in V.

(2) Associativity: For all v_1, v_2 and v_3 in V, $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$.

(3) Additive identity: There exists an element 0 in V such that, for every element v in V, the equation $0 + v = v + 0 = v$ holds.

(4) Additive Inverse: For each v in V, there exists an inverse element $-v$ in V such that $v + (-v) = (-v) + v = 0$.

(5) Multiplicative Identity in Field F for Scalar Multiplication: For each v in V, $1 \cdot v = v$.

(6) Compatibility of Scalar Multiplication with Field Multiplication: For any a_1 and a_2 in F and each v in V, $a_1 \cdot (a_2 \cdot v) = (a_1 \times a_2) \cdot v$.

(7) Distributivity of Scalar Multiplication with respect to Vector Addition: For any a in F and each v_1 and v_2 in V, $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$.

(8) Distributivity of Scalar Multiplication with respect to Field Addition: For any a_1 and a_2 in F and each v in V, $(a_1 + a_2) \cdot v = a_1 \cdot v + a_2 \cdot v$.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

结合原文理解：实数完备性是多条定理背后的共同原因。做证明题时，尤其注意“非空、有界、闭、收敛”的区别。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.54 第 54 页｜本页课件

Distance - Metric

N-dimensional Euclidean space, or N-dimensional continuum, is the set $\mathbb{R}^N$ of all number tuple of N real numbers.

$$\mathbb{R}^N = \{x = (x_1, \cdots, x_N) : x_n \in \mathbb{R} \text{ for } n = 1, \cdots N\}.$$

For any two points $x^i = (x_1^i, \cdots, x_N^i)) \in \mathbb{R}^N$ for $i = 1, 2$, define the distance between the two points

$$\rho(x^1, x^2) = \left\{ \sum_{n=1}^{N} \left(x_n^1 - x_n^2\right)^2 \right\}^{\frac{1}{2}} = \sqrt{\sum_{n=1}^{N} (x_n^1 - x_n^2)^2}.$$

Then we have the following three propositions.

(1) Positivity: $\rho(x^1, x^2) \geqslant 0$. $\rho(x^1, x^2) = 0$ if and only if $x^1 = x^2$.

(2) Symmetry: $\rho(x^1, x^2) = \rho(x^2, x^1)$.

(3) Triangle Inequality: $\rho(x^1, x^3) \leqslant \rho(x^1, x^2) + \rho(x^2, x^3)$.

From the triangle inequality, $\rho(x^1, \cdot)$ is a continuous function on x_2:

$$\lim_{\rho(x_n^2, x^2) \to 0} \rho(x^1, x_n^2) = \rho(x^1, x^2).$$

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

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读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.55 第 55 页｜ Norm

Norm

N-dimensional Euclidean space, or N-dimensional continuum, is the set $\mathbb{R}^N$ of all number tuple of N real numbers.

$$\mathbb{R}^N = \{x = (x_1, \cdots, x_N) : x_n \in \mathbb{R} \text{ for } n = 1, \cdots N\}.$$

For any two points $x = (x_1, \cdots, x_N)) \in \mathbb{R}^N$, define the norm as

$$\|x\| = \left\{\sum_{n=1}^{N} x_n^2\right\}^{\frac{1}{2}} = \sqrt{\sum_{n=1}^{N} x_n^2} = \rho(x, 0).$$

Then we have the following three propositions.

(1) Positivity: $\|x\| \geqslant 0$. $\|x\| = 0$ if and only if $x = 0$.

(2) Symmetry: $\|x\| = \| - x\|$.

(3) Triangle Inequality: $\|x^1 + x^2\| \leqslant \|x^1\| + \|x^2\|$.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

结合原文理解：实数完备性是多条定理背后的共同原因。做证明题时，尤其注意“非空、有界、闭、收敛”的区别。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.56 第 56 页｜Theorem on Minimum and Maximum

Theorem on Minimum and Maximum

Theorem on Minimum and Maximum Suppose E is a non-empty bounded set in $\mathbb{R}^N$. Then E has minimum and maximum members.

Pre-ordering: Functional Analysis

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：最大值定理的逻辑：非空紧值、连续可行对应 + 连续目标推出值函数连续，argmax 对应非空紧值且上半连续。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.57 第 57 页｜Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence

Theorem on Limit of Bounded Monotone Sequence Every bounded monotone sequence in $\mathbb{R}$ has a limit in $\mathbb{R}$.

Theorem of Limit of Increasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is an increasing sequence of real numbers which is bounded above. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Theorem of Limit of Decreasing Sequence Suppose $\{x_n : n \in \mathbb{N}\}$ is a decreasing sequence of real numbers which is bounded below. Then there exists a limit of $\{x_n : n \in \mathbb{N}\}$, or $\lim_{n\to\infty} x_n$ exists.

Ordered Set: Mathematical Analysis

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

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2.58 第 58 页｜ Cantor Nested Intervals Theorem

Cantor Nested Intervals Theorem

Nested Intervals Theorem If the interval $\{I_n : n \in \mathbb{N}\}$ is a sequence of closed bounded nested intervals in $\mathbb{R}^N$ and $\lim_{n\to\infty} |I_n| = 0$, then there exists a point $\xi = \lim_{n\to\infty} I_n \in \bigcap_{n=1}^{\infty} I_n$.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「nested」证明专题。

2.59 第 59 页｜Bolzano Theorem

Bolzano Theorem

Bolzano Theorem Each bounded sequence in $\mathbb{R}^N$ has a convergent subsequence.

Bolzano Theorem Each bounded sequence in $\mathbb{R}^N$ has a limit point.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：有界数列存在收敛子列。反复把区间一分为二，始终留下包含无穷多项的半区间。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「bw」证明专题。

2.60 第 60 页｜Weierstrass Theorem on Accumulation Point

Weierstrass Theorem on Accumulation Point

Weierstrass Theorem Every bounded infinite set in $\mathbb{R}^N$ has an accumulation point.

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

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对应的完整证明：参见本讲义最后的「bw」证明专题。

2.61 第 61 页｜ Cauchy Convergence Criterion

Cauchy Convergence Criterion

Cauchy Convergence Criterion The sequence $\{x_n : n \in \mathbb{N}\}$ in $\mathbb{R}^N$ is convergent if and only if it is a Cauchy sequence: for any $\varepsilon > 0$, there exist a large natural number K such that, when $n_1 > K$ and $n_2 > K$, $\|x_{n_1} - x_{n_2}\| < \varepsilon$.

Completeness: Functional Analysis

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：柯西条件只比较尾部的各项，而不预设极限；其充分性使用实数的完备性。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「cauchy」证明专题。

2.62 第 62 页｜ Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem on Finite Coverings

Heine - Borel Theorem If $\mathscr{H}$ is an open covering of a closed and bounded subset E of $\mathbb{R}^N$; then E has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Compactness: Functional Analysis

这一页怎么读

概念主线：本章从实数的有序域公理与最小上界性质出发，建立数列收敛、紧致性、连续性所需的基础。

本页重点：开覆盖具有有限子覆盖的结论必须依赖紧致性。证明时可用闭区间套反证或逐次细分。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

对应的完整证明：参见本讲义最后的「hb」证明专题。

3 证明专题：考试应会的标准推导

专题的使用方法

这些专题沿着本课 PPT 的定理顺序补足主要推导；若原课件的命题另有附加假设，以该页原文为主，切勿将证明中的条件删掉。对于高维拓扑不动点定理，只陈述严格的适用条件并说明所需工具，不伪称给出了没有依据的完整高维证明。

3.1 专题：complete

证明｜为什么有理数不是完备的

考虑 $A=\{q\in\mathbb{Q}:q>0,\ q^2<2\}$。它在 $\mathbb{Q}$ 内非空、有上界，却没有有理数上确界。若 $s\in\mathbb{Q}$ 是其上确界，则 $s^2=2$ 才可能；若 $s^2<2$，可以在 s 附近找更大的有理数仍满足平方小于 2；若 $s^2>2$，可以找一个稍小的有理数仍是上界。这两种情况均与上确界矛盾。又由初等数论 $\sqrt{2}\notin\mathbb{Q}$（假设 $\sqrt{2}=a/b$ 为最简分数，可推出 a,b 均为偶数），故 $\mathbb{Q}$ 不完备。

3.2 专题：supremum

证明｜上确界的 ε 刻画

设 $A\subset\mathbb{R}$ 非空且有上界，$s=\sup A$。第一，由定义，对任意 $a\in A$ 有 $a\le s$。第二，对任意 $\varepsilon>0$，若不存在 $a>s-\varepsilon$，则 $s-\varepsilon$ 也是上界，与 s 为最小上界矛盾。所以 $\exists a\in A$ 使 $s-\varepsilon<a\le s$。反过来，若 s 是上界且满足上述逼近性，假设有更小上界 $b<s$，取 $\varepsilon=s-b$，找到 $a>b$，矛盾。

3.3 专题：archimedean

证明｜由最小上界性质证明阿基米德性

任取 $x\in\mathbb{R}$。若对每个 $n\in\mathbb{N}$ 都有 $n\le x$，则 $\mathbb{N}$ 有上界，令 $s=\sup\mathbb{N}$。因为 $s-1$ 不是上界，存在 $n>s-1$，于是 $n+1>s$。但 $n+1\in\mathbb{N}$，与 s 是上界矛盾。故存在 $n\in\mathbb{N}$ 使 $n>x$。

3.4 专题：nested

证明｜闭区间套定理（完整量词）

设 $I_n=[a_n,b_n]$ 为一列非空闭区间，$I_{n+1}\subseteq I_n$。则 a_n 单调不减、b_n 单调不增，且 $a_n\le b_m$ 对所有 m,n 成立（取足够大的 $k\ge m,n$ 可得 $a_n\le a_k\le b_k\le b_m$）。集合 $A=\{a_n:n\ge1\}$ 非空、有上界，令 $c=\sup A$。对任意固定 m，b_m 是 A 的上界，故 $c\le b_m$；同时 $a_m\le c$。因此 $a_m\le c\le b_m$，从而 $c\in\bigcap_n I_n$。若再有 $b_n-a_n\to0$，任意 c,d 在交集中都满足 $|c-d|\le b_n-a_n\to0$，故 $c=d$，交集恰有一个点。

3.5 专题：bw

证明｜Bolzano–Weierstrass：有界数列存在收敛子列

一维时，取包含数列所有项的闭区间 I_0。把它平分为左右两个闭区间，至少有一个包含无穷多个数列项；选这样的半区间 I_1。反复操作得嵌套 I_k，长度为 $|I_0|/2^k \to 0$。由闭区间套定理 $\bigcap_k I_k = \{c\}$。依次选严格递增的下标 n_k，使 $x_{n_k} \in I_k$。于是 $|x_{n_k} - c| \le |I_k| \to 0$。在 $\mathbb{R}^N$，可逐坐标抽取子列，共进行有限次，得到所有坐标都收敛的共同子列。

3.6 专题：cauchy

证明｜柯西收敛准则

必要性：若 $x_n \to x$，给定 $\varepsilon > 0$，找 N 使 $n > N$ 时 $|x_n - x| < \varepsilon/2$，则 $m, n > N$ 时 $|x_m - x_n| \le |x_m - x| + |x_n - x| < \varepsilon$。充分性：柯西列有界（尾部各项与某固定项距离小于 1，再并入前面有限项）；由 Bolzano–Weierstrass 定理有收敛子列 $x_{n_k} \to x_\circ$。给定 ε，先用柯西性令尾部项彼此距离 $< \varepsilon/2$，再选足够大的 n_k 使 $|x_{n_k} - x| < \varepsilon/2$，于是所有足够大的 n 满足 $|x_n - x| < \varepsilon$。

3.7 专题：hb

证明｜Heine–Borel：闭且有界集的每个开覆盖有有限子覆盖

先证明闭立方体 $Q \subset \mathbb{R}^N$ 的结论。反设某开覆盖没有有限子覆盖。将 Q 等分成 2^N 个小立方体，至少有一个仍无法有限覆盖；递归选择这样的嵌套闭立方体 Q_k，其直径趋零。由多维闭区间套，存在公共点 $c_\circ$。开覆盖中有一个开集 U 含 c，故存在 $r > 0$ 使 $B(c, r) \subset U$；当 k 足够大时 $Q_k \subset B(c, r)$，于是 Q_k 可用一个开集覆盖，矛盾。若 F 闭且有界，把它放入大立方体 Q；将 F 的开覆盖加上 $\mathbb{R}^N \setminus F$，构成 Q 的开覆盖并取有限子覆盖，去掉补集即可。

3.8 专题：ivt

证明｜介值定理（以零点为例）

设 $f : [a, b] \to \mathbb{R}$ 连续，$f(a) < 0 < f(b)$。令 $A = \{x \in [a, b] : f(x) \le 0\}$，因 $a \in A$，A 非空且有上界 b，设 $c = \sup A_\circ$。若 $f(c) < 0$，连续性保证 c 右侧足够近的点仍使 $f < 0$，违背 c 为上界。若 $f(c) > 0$，连续性保证 c 左侧足够近的点都使 $f > 0$，违背 c 是 A 的最小上界（用上确界的 ε 刻画）。因此 $f(c) = 0_\circ$ 一般介值 v 取 $g(x) = f(x) - v$ 即可。

3.9 专题：evt

证明｜连续函数在非空紧集上取得最大、最小值

设 $K \subset \mathbb{R}^N$ 非空紧致，$f : K \to \mathbb{R}$ 连续。先证有界：若无上界，取 $x_n \in K$ 使 $f(x_n) > n_\circ$ 紧致性给出收敛子列 $x_{n_k} \to x \in K$，连续性给出 $f(x_{n_k}) \to f(x)$，与 $f(x_{n_k}) > n_k \to \infty$ 矛盾。令 $M = \sup_{x \in K} f(x)$，依上确界性质取 $x_n \in K$ 使 $f(x_n) > M - 1/n$，取收敛子列 $x_{n_k} \to x^* \in K$，连续性推出 $f(x^*) = M_\circ$ 对 $-f$ 重复论证即得最小值。

3.10 专题：compact

证明｜为什么有限维的“闭且有界”能推出序列紧致

设 $K \subset \mathbb{R}^N$ 闭且有界，任取 $x_n \in K$。因有界，Bolzano–Weierstrass 定理保证存在子列 $x_{n_k} \to x \in \mathbb{R}^N$；因 K 闭，极限 $x \in K$。这就证明了 K 的每个序列都有极限仍在 K 的收敛子列。反之若 K 不有界，可以选 $\|x_n\| > n$，不可能有收敛子列；若不闭，则存在 K 内的收敛序列趋向外部点，序列紧致性与子列唯一极限矛盾。

4 自测清单

1. 你能不看课件写出本讲的核心定义及所有量词吗?
2. 你能逐条列出每一个定理的假设，并解释删去一条为何可能失败吗?
3. 你能在白纸上独立复述本讲证明专题中的每一个箭头吗?
4. 你能结合原课件的反例判断是否把充分条件错写成必要条件吗?